Interactions, Inclusion–Exclusion, and Partition Functions

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One idea carries this whole guide: multiply together a set of "keep this part / throw this part away" operators, expand the product, and the alternating signs fall out. Möbius inversion, inclusion–exclusion, ANOVA decomposition and your coefficient-recovery formula are all that one expansion. The partition-function picture in Step 6 then explains what interaction overlap means at the portfolio level.

What you need to know first

None of these needs lattices, posets or category theory. Items marked you already have from physics.

Concept	What you need from it	Where it's used
Sets and subsets	$S \subseteq T$, union, intersection, set difference $T \setminus S$, size $\lvert S\rvert$, "sum over all subsets U of S" (there are $2^{\lvert S\rvert}$ of them)	Steps 2–4
Expanding a product of sums	$(a+b)(c+d) = ac+ad+bc+bd$: one choice from each factor per term	Step 3
Linear operators	Composition, the identity I, when two operators commute	Steps 3–4
Projections	An operator with $PP = P$. Then $Q = I - P$ is also one, and $PQ = 0$	Steps 3–5
Finite differences	$f(x+h) - f(x)$; the mixed version is the discrete $\partial^2 f/\partial x \partial y$	Steps 1, 4
Weighted averages	$\langle f\rangle_w = \sum wf / \sum w$, and changing the weights	Step 6
Kronecker delta, Levi-Civita, permutation parity	As in tensor calculus	Step 5
Partition function, free energy, cumulant expansion	$Z = \sum e^{-\beta E}$, $F = -kT \ln Z$, $\ln\langle e^X\rangle = \langle X\rangle + \mathrm{Var}(X)/2 + \cdots$	Step 6

Two terms get defined in plain words when they appear: **Möbius inversion** (undoing a running total over subsets) and **anchored vs. averaged decomposition** (two choices of projection).

Step 1: The one-dimensional seed

A running total and a difference undo each other. That is the discrete fundamental theorem of calculus, and every alternating sum in this guide is a many-variable version of it.

Suppose $g(n)$ is built by stacking pieces $f(m)$:

$$g(n) = \sum_{m \le n} f(m)$$

To recover a single piece, subtract the total just below it:

$$f(n) = g(n) - g(n-1)$$

There is one minus sign because there is exactly one "level below" n. In Step 2, "below" becomes "is a subset of", and a set has many subsets just below it. That's the only reason the signs multiply.

Physical reading. g is a cumulative quantity (total charge enclosed, a CDF). f is the density. Differencing strips away everything accumulated before, leaving only what's new at n.

Step 2: Many variables — running totals over subsets

Möbius inversion is Step 1 with "$m \le n$" replaced by "$U \subseteq S$". Its alternating signs come from one fact: $(1-1)^k = 0$.

Building up. Index pieces by sets of variables. A total over S stacks every piece belonging to a subset of S:

$$g(S) = \sum_{U \subseteq S} f(U)$$

With two variables A and B, written out:

Total	Pieces it contains
$g(\emptyset)$	$f(\emptyset)$
$g(\{A\})$	$f(\emptyset) + f(A)$
$g(\{B\})$	$f(\emptyset) + f(B)$
$g(\{A, B\})$	$f(\emptyset) + f(A) + f(B) + f(AB)$

Peeling back. To isolate $f(AB)$, take the top total, subtract the two below it, then add back $f(\emptyset)$, which was subtracted twice:

$$f(AB) = g(\{A, B\}) - g(\{A\}) - g(\{B\}) + g(\emptyset)$$

This is the Venn-diagram rule $|A \cup B| = |A| + |B| - |A \cap B|$ run in reverse. In general:

$$f(S) = \sum_{U \subseteq S} (-1)^{|S|-|U|}\, g(U)$$

That is **Möbius inversion** on subsets. The sign is + for subsets missing an even number of S's variables and − for an odd number.

Why it works, in one line. Pick any piece $f(V)$ with V a strict subset of S. It appears in $g(U)$ for every U between V and S. There are 2^k such U, where $k = |S \setminus V|$, and their signs sum to $(1-1)^k = 0$. So every lower-order piece cancels and only $f(S)$ survives.

Three variables, to see the pattern grow: add the three two-variable totals, subtract the three one-variable totals, add $g(\emptyset)$, all with the right parity.

$$f(ABC) = g_{ABC} - g_{AB} - g_{AC} - g_{BC} + g_A + g_B + g_C - g_\emptyset$$

The coefficient-recovery formula from earlier is exactly this, with \$g(U) = \$ log of the two-way partial dependence with the variables in U at their chosen levels and the rest of S at base. Step 4 shows why $\ln \hat{u}$ plays the role of the running total.

Step 3: The engine — a product of commuting projections

Give each variable k an operator P_k that "forgets" x_k. Writing the identity as a product of (forget + remainder) and expanding it decomposes any function into pure pieces, one per subset of variables. The signs of Step 2 appear the moment you expand the remainders.

The ingredients. For each variable k, choose an operator P_k on functions of x such that:

1. $P_k f$ no longer depends on x_k (it forgets x_k),
2. $P_k P_k = P_k$ (forgetting twice is forgetting once, so P_k is a projection),
3. $P_j P_k = P_k P_j$ (forgetting in either order gives the same result).

Let $Q_k = I - P_k$, "what forgetting x_k throws away". Then $P_k Q_k = 0$.

The identity. Since $P_k + Q_k = I$ for each k, the product over all p variables is also I. Expand it as in $(a+b)(c+d)$: each term picks P or Q from every factor.

$$I = \prod_{k=1}^{p} (P_k + Q_k) = \sum_S \Big(\prod_{k \in S} Q_k \Big) \Big(\prod_{k \notin S} P_k \Big)$$

Apply it to f and name each term:

$$f = \sum_S f_S, \qquad f_S = \Big(\prod_{k\in S} Q_k\Big)\Big(\prod_{k\notin S} P_k\Big) f$$

What each piece is.

- f_S depends only on the variables in S, because every other variable was forgotten.
- f_S is **pure**: $P_k f_S = 0$ for every k in S, because $P_k Q_k = 0$. Nothing in f_S survives forgetting any one of its variables, so none of it could have been explained by fewer variables.
- The pieces are unique once the P_k are chosen.

Where the signs come from. Substitute $Q_k = I - P_k$ inside f_S and expand again:

$$f_S = \sum_{U\subseteq S} (-1)^{|S|-|U|}\, g(U), \qquad g(U) = \Big(\prod_{k\notin U} P_k\Big) f$$

$g(U)$ is f with everything outside U forgotten, and it equals the sum of the pieces f_V for $V \subseteq U$. So g is the running total of Step 2, and the formula is Möbius inversion. You never need to invoke Möbius inversion by name: it is what $(I - P)$ expands into.

Physical reading. P_k coarse-grains x_k away. Q_k keeps only the fluctuation in x_k. The interaction f_S is the part of f that fluctuates jointly in all of S and in no smaller set.

A caution on the text you pasted: the formula $f_{12} = (I - P_1)(I - P_2)f$ is complete only when f depends on x_1 and x_2 alone. With more variables, the others must be forgotten too, i.e. multiplied by $P_3 P_4 \cdots$, exactly as the f_S formula above says.

Step 4: Choosing the projection — anchored vs. averaged

Corner-point coefficients and ANOVA components are the same Step 3 machine with different P_k. The mixed differences come out identical either way. Only the split into "main effect" and "interaction" depends on the choice.

Anchored (corner-point). Forget x_k by setting it to its base level. In your operator notation:

$$(P_k f)(x) = f(\mathbb{T}_k(z_0)\, x), \qquad (Q_k f)(x) = f(x) - f(\mathbb{T}_k(z_0)\, x)$$

Q_k is the difference against base, the Δ_k of earlier. Apply Step 3 to $f = \ln \hat{u}$. The pieces f_S are exactly the corner-point β_S, and $f_S = Q_A Q_B \ln \hat{u}$ is the mixed difference. The coefficient-recovery formula is this expansion.

Averaged (ANOVA / Hoeffding). Forget x_k by averaging it over its exposure mix π_k:

$$(P_k f)(x) = \sum_z \pi_k(z)\, f(\mathbb{T}_k(z)\, x)$$

For one record, that is the mix-weighted average of its ICE curve. The pieces are centered: each averages to zero over its own variables, like sum-to-zero coding. mgcv's centering of smooths is this kind of choice (it sums to zero over the data); a `bs = "re"` term is neither, since its split is set by the penalty.

Worked example. Two binary factors, equal weights, four cells of $\ln \widehat{u}$:

Cell	$\ln \widehat{u}$	Anchored: $\emptyset + A + B + AB$	Averaged: $\emptyset + A + B + AB$
a_0, b_0	1.0	$1.0 + 0 + 0 + 0$	$1.15 - 0.25 + 0 + 0.10$
a_1, b_0	1.3	$1.0 + 0.3 + 0 + 0$	$1.15 + 0.25 + 0 - 0.10$
a_0, b_1	0.8	$1.0 + 0 - 0.2 + 0$	$1.15 - 0.25 + 0 - 0.10$
a_1, b_1	1.5	$1.0 + 0.3 - 0.2 + 0.4$	$1.15 + 0.25 + 0 + 0.10$

What the table shows:

- **The interaction contrast is the same.** Anchored: $0.4 - 0 - 0 + 0 = 0.4$. Averaged: $0.10 + 0.10 + 0.10 + 0.10 = 0.4$. Differencing in every variable of S kills anything not depending on all of S, and both choices differ only in such lower-order pieces.
- **The main effects are not.** B's main effect is -0.2 anchored and exactly 0 averaged. Anchored reports B's effect at $A = a_0$. Averaged reports it averaged over A. Neither is wrong; they answer different questions.
- **The A contrast is 0.3 vs. 0.5.** The effect of A is 0.3 at b_0 and 0.7 at b_1. Anchored reports the first; averaged reports their mean.

The takeaway. "Pure interaction" is well defined only as a fully differenced quantity. Assigning an interaction's mass to cells, or calling something a main effect, requires choosing what "forgetting a variable" means. That choice is the coding.

Step 5: Kronecker delta and Levi-Civita — what the connection really is

The Kronecker delta connection is exact: it's a basis expansion. The Levi-Civita connection is a genuine analogy (both build a projection and expand it into signed terms) but the signs come from different places. The cleanest geometric link is a different one: the interaction contrast is the integrated twist of the surface $\ln \widehat{u}$.

Kronecker delta: exact. The indicators $\delta(\ell, x_k)$ over the levels ℓ of variable k form a complete set: for any record, exactly one is 1, so they sum to 1. The lookup is an expansion in that set, and an interaction expands in products of indicators:

$$\beta_k(x_k) = \sum_{\ell} \beta_k(\ell)\, \delta_{\ell,x_k}, \qquad \beta_{jk}(x_j, x_k) = \sum_{\ell,m} \beta_{jk}(\ell, m)\, \delta_{\ell,x_j}\, \delta_{m,x_k}$$

In this picture, a coding is a choice of basis. Corner-point drops the base level's indicator; sum-to-zero uses contrast vectors. The projections of Step 4 act on the coefficient tables the same way they act on functions.

Levi-Civita: an analogy. For two indices, let S swap them. The antisymmetrizer $A = (I - S)/2$ satisfies $AA = A$, so it is a projection, and its complement $I - A$ is the symmetrizer. Written out in components, A is the difference of products of deltas, which is what the ε-δ identity expresses:

$$\epsilon_{kij}\, \epsilon_{kmn} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$$

So both constructions do the same thing: build a projection onto a "pure" part and expand it, producing signed terms. They differ in the details:

	ANOVA / coefficient recovery	Antisymmetrization
Building blocks	One projection P_k per variable	One projection built from all index permutations
Do the pieces commute?	Yes	No (permutations generally don't)
Terms for 3 variables / indices	8 (subsets of 3)	6 (permutations of 3)
Sign of a term	(-1) to the number of variables forgotten	Parity of the permutation

For two variables both patterns have one plus and one minus, which is why they look alike. From three on they diverge. The pasted summary's claim that Levi-Civita is Möbius inversion on a lattice of subspaces overstates it: the permutation signs come from permutation parity, not from inclusion–exclusion.

The geometric link that is exact. The mixed difference over a rectangle of levels is the integral of the mixed partial derivative over that rectangle. The four corners pick up alternating signs, exactly as in Stokes' theorem on a square:

$$f(a,b) - f(a,b_0) - f(a_0,b) + f(a_0,b_0) = \int_{a_0}^{a}\int_{b_0}^{b} \frac{\partial^2 f}{\partial x\, \partial y}\, dy\, dx$$

With $f = \ln \hat{u}$, an additive model is a surface with $\partial^2 f / \partial x \partial y = 0$ everywhere, so it has no twist. An interaction coefficient measures how much the log-loss-cost surface twists between the base corner and cell (a, b). For categorical variables the integral becomes the four-corner sum, but the meaning is the same.

Step 6: The portfolio level — partition functions

A weighted sum of $\hat{u}$ over the book is a partition function, with $\ln \hat{u}$ playing the role of minus the energy. Forcing variables clamps a subsystem. A coefficient is an energy difference; a PDP relativity is a free-energy difference. They coincide exactly when the clamped subsystem is decoupled from the rest, which is the closure condition. When it isn't, the bridge identity is the free-energy perturbation formula.

The dictionary (take $kT = 1$):

Insurance	Statistical mechanics
Record i	Microstate
Exposure e_i	Degeneracy / prior weight of the microstate
$\ln \hat{u}(x_i)$	$-E_i$, minus the energy
$\Sigma_e(\hat{u})\, D$	Partition function Z
Forcing $\mathbb{T}_C(z)$	Clamping subsystem C into configuration z
Coefficient difference, e.g. $\beta_k(z) - \beta_k(z_0)$	$-\Delta E$ of the subsystem's own energy ("bare")
ln of the PDP relativity	$-\Delta F$, a free-energy difference ("dressed")
Interaction term straddling C and the rest	Coupling energy between subsystem and bath
Closed set C	Decoupled subsystem
One-way via $\mathbb{F}$	A different ensemble (a sub-population), not a clamped one

Decoupled: Z factorizes. If the energy splits as $E = E_C(z) + E_{\text{rest}}(x)$ with no coupling term, then clamping C multiplies Z by a constant:

$$Z(z) = e^{-E_C(z)}\, Z_{\text{rest}} \quad \Longrightarrow \quad \ln \frac{Z(z)}{Z(z_0)} = -[E_C(z) - E_C(z_0)]$$

That is the factoring-out step from the start of our discussion. Free energy equals bare energy, so the relativity equals the coefficient.

Coupled: free-energy perturbation. Let X_i be record i's log ICE relativity. The bridge identity is Zwanzig's formula, with the average taken in the reference ensemble (everything clamped at z_0, weights $e \cdot \hat{u} \circ \mathbb{T}(z_0)$):

$$\frac{Z(z)}{Z(z_0)} = \langle e^X \rangle_0 \quad \Longrightarrow \quad \ln(\text{PDP relativity}) = \langle X \rangle_0 + \tfrac{1}{2}\mathrm{Var}_0(X) + \cdots$$

$X_i = _k + $ (the coupling terms evaluated at record i's bath variables). So:

- **Mean term.** $\langle X \rangle_0$ is the bare coefficient plus the coupling averaged over the reference ensemble.
- **Fluctuation term.** If the coupling varies across records, the variance term adds to the relativity. By Jensen's inequality, $\ln(\text{PDP relativity}) \geq \langle X \rangle_0$, with equality only when X is the same for every record.
- **Link-scale PD** averages X under plain exposure weights. That is a mean energy in an untilted ensemble, which is why it differs from the response-scale PD.

Zwanzig's identity is the instantaneous-switching case of Jarzynski's equality, so the PDP relativity is literally a free-energy difference for an instantaneous "switch" of every record's level.

What "interaction overlap" means. A coefficient is the bare energy of the variables you clamped. A partial relativity is that energy dressed by coupling to the rest of the book. The terms that straddle the boundary of the clamped set are the couplings. Clamping every coupled partner (closure) removes the bath's influence, and the dressed and bare quantities coincide. A one-way goes further: it changes the ensemble itself, so it also mixes in differences in the bath's composition.

Summary and further reading

One structure seen four ways:

Idea	Combinatorics	Operators	Geometry	Physics
Building up	Running total over subsets	Forgetting variables: $\prod P_k$	—	Bath at equilibrium
Isolating a piece	Möbius inversion, signs $(-1)^{\lvert S\rvert-\lvert U\rvert}$	Expanding $\prod(I - P_k)$	Mixed difference = integrated twist	Bare energy of a subsystem
Why lower orders cancel	$(1-1)^k = 0$	$P_k Q_k = 0$	Flat surfaces have no twist	Decoupled terms factor out of Z
Choice that remains	Which totals count as "below"	Anchored vs. averaged P_k	Which corner is the base	Reference ensemble
Portfolio quantity	—	$\mathbb{T}$ then Σ_e	—	Free energy (dressed)

Further reading. These are standard references cited from memory, not looked up; titles and years are approximate.

- Rota (1964), "On the foundations of combinatorial theory I: Theory of Möbius functions": the general form of Step 2.
- Hoeffding (1948): the averaged decomposition of Step 4.
- Rabitz and Aliş (1999) on high-dimensional model representation, including the anchored ("cut") version: the corner-point decomposition of Step 4.
- Zwanzig (1954) on free-energy perturbation and Jarzynski (1997) on its nonequilibrium generalization: Step 6.